ANNAPURNA KALMATH

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Prompt robustness Analyzer [*Live Link*][*GitHub Link*]

Independent | July 2026 – Present

- Built and shipped an **11-stage prompt** diagnostics platform that analyzes, stress-tests, repairs, and regression-validates AI prompts for specification completeness before deployment.
- Represents prompts as a validated tree of **persona, goals, tasks, constraints, and reasoning directives,** enabling structural analysis for contradictions, redundant instructions, scope conflicts, and missing specifications.
- Generates realistic **non-adversarial edge cases** across 12 failure categories, executes them against the prompt, consolidates related failures into generalized fixes, produces a unified prompt rewrite, and regression-tests it against the same evaluation suite.
- Supports both **Analyze** (audit existing prompts) and **Build** (generate prompts directly from structured specifications), treating prompts as executable specifications rather than free-form text.
- Produces specification-aware rewrites that automatically expand underspecified prompts and compress redundant ones, with every rewrite regression-tested against generated obstacle cases.

Founder Memory OS [*Video Demo*]

Independent | May 2026 – Present

- Built an event-driven AI memory platform that transforms founder conversations, voice notes, and documents into a continuously evolving technical memory, preserving context across product development.
- Developed an AWS-based ingestion pipeline that extracts architectural decisions, abandoned approaches, contradictions, and unexplored technical opportunities from unstructured conversations.
- Built a **conversation archaeology engine** that reconstructs product evolution into a versioned knowledge base, enabling contextual recall, knowledge navigation, gap discovery, teaching, and technical reasoning instead of transcript search.
- Designed for **long-term product memory rather than chat history**, treating product development as an evolving knowledge graph instead of isolated conversations.

Organisational Knowledge Infrastructure [*Product Memo Link*]

Independent · | May 2026 – Present

- Built a continuously evolving **organizational knowledge infrastructure** that transforms fragmented company documents into structured, traceable knowledge for AI agents and human decision-making.
- Developed a **two-layer knowledge graph** separating official organizational knowledge (policies, workflows, decisions, commitments) from operational reality (incidents, support activity, observations), enabling contradiction detection, evolution tracing, and organizational drift analysis.
- Represented organizational knowledge as **typed, traceable claims** rather than documents, preserving governance relationships, policy evolution, cross-functional dependencies, and full provenance through graph construction and retrieval.
- Built an end-to-end prototype that automatically extracts claims, constructs typed relationships, and supports evidence-backed reasoning over a graph with **130+ nodes and 160+ relationships** generated from a synthetic enterprise corpus.

Legal Research Infrastructure [*technical video demo: Link*] [*Product paper: Link*]

Nov 2025 – Feb 2026

- Built a **stateful legal reasoning engine** that operates **after legal retrieval**, supporting litigation strategy by reasoning over case facts and judicial decisions instead of returning search results.
- Designed an **IRAC-based judgment representation** and **fact-to-precedent alignment pipeline** that retrieves judicial reasoning units, maps client facts to precedent, and adapts analysis to each judgment's doctrinal framework rather than relying on semantic similarity.
- Implemented **citation-grounded reasoning workflows** for precedent auditing, defense generation, and litigation risk analysis, with every conclusion linked to the exact supporting passages from user-uploaded judgment packs for transparent verification.
- Validated the end-to-end prototype with practicing lawyers using real judgment packs, demonstrating the system's ability to surface weak precedents, doctrinal conflicts, and strategic arguments before pausing development due to domain-specific expertise requirements.

RESEARCH

Recommender Systems as Neurobehavioral Infrastructures (Link to SSRN)

- Authored an interdisciplinary conceptual synthesis connecting large-scale recommendation system architectures with cognitive neuroscience, proposing a framework for analyzing how ranking objectives interact with reward learning, habit formation, and developmental vulnerability.

Education:

B.E. Electronics & Communication Engineering- SDM College of Engineering & Technology | CGPA: 9.69/10 (Top 3% of class)
GATE 2025 (Data Science & AI): AIR 1911 (Top 3% nationally)
Admitted to **IIIT Hyderabad (MS by Research)**; withdrew to build full-time
Built and killed multiple MVPs based on customer interviews and market validation.

Skills:

Python, Java | LLM Systems, RAG, Multi-Agent Systems | Knowledge Graphs, Information Retrieval, Knowledge Representation | AWS, System Design